

Middle Convolution of ℓ -adic Perverse Sheaves

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Contents

1	Introduction	2
1.1	Notations	2
1.2	Motivations	3
1.3	Structure of the Write-up	3
1.4	Acknowledgements	4
2	Local Systems and Lisse Sheaves	4
3	Perverse Sheaves and the Six Functor Formalism	4
4	Middle Convolution and Monodromy Groups	4
5	Mod-ℓ Monodromy of Hyperelliptic Curves	4
6	Formalization in Lean	4

1 Introduction

1.1 Notations

1.1.1. Motivated by the growing interest in the formalization of mathematics within the mathematical community, we adopt a style of notation inspired by type theory throughout this write-up, without committing to type-theoretic foundations (foundational questions lie outside the scope of this write-up). The symbols below denote types and/or categories depending on your philosophical inclination, where $X : T$ denotes that X is a term of type T , or an object of the category T .

Set	Sets
Set^{fin}	Finite sets
Nat	Natural numbers (including zero)
Int	Integers
Ab	Abelian groups
Ring	Rings (associative, unital, not necessarily commutative)
Field	Fields
Mod_R	Left modules over $R : \text{Ring}$
Vect_k	Vector spaces over $k : \text{Field}$
$\text{Vect}_k^{\text{fin}}$	Finite-dimensional vector spaces over $k : \text{Field}$
Sch	Schemes
Sch_S	Schemes over base scheme $S : \text{Sch}$
Var_k	Varieties over $k : \text{Field}$, i.e. integral separated schemes of finite type over k
Curv_k	Curves over $k : \text{Field}$, i.e. one-dimensional varieties over k
$\text{Curv}_k^{\text{sm}}$	Smooth curves over $k : \text{Field}$
Riem	Riemann surfaces, i.e. connected one-dimensional complex manifolds
Site	Sites, i.e. categories equipped with Grothendieck topologies
$\text{Sh}(X, A)$	Sheaves on $X : \text{Site}$ with values in a category A admitting small limits
$\text{Mod}(X, \mathcal{R})$	Sheaves of left $\mathcal{R}$ -modules on $X : \text{Site}$, where $\mathcal{R} : \text{Sh}(X, \text{Ring})$
$D^b(A)$	Bounded derived objects of an abelian category A
$D^b(X, k)$	Bounded derived objects of $\text{Sh}(X, \text{Vect}_k)$, where $X : \text{Site}$ and $k : \text{Field}$
$X \rightarrow Y$	Morphisms from $X : T$ to $Y : T$

1.1.2 Global Notations. The following terms will be used throughout the write-up

$(-)^{\text{an}} : \text{Curv}_{\mathbb{C}}^{\text{sm}} \rightarrow \text{Riem}$	The complex analytification functor
$[n] : D^b(X, k) \rightarrow D^b(X, k)$	Cohomological shift by $n : \text{Int}$, for $X : \text{Site}$
$\Gamma(X, -) : D^b(X, k) \rightarrow D^b(\text{Vect}_k)$	Derived global sections for $X : \text{Site}$
$\mathcal{H}^i : D^b(X, k) \rightarrow \text{Sh}(X, \text{Vect}_k)$	The i -th cohomology sheaf, for $i : \text{Int}$ and $X : \text{Site}$
$H^i(X, -) : D^b(X, k) \rightarrow \text{Vect}_k$	The i -th hypercohomology, for $i : \text{Int}$ and $X : \text{Site}$
$\underline{V}_X : \text{Sh}(X, \text{Vect}_k)$	The constant sheaf associated with $V : \text{Vect}_k$ on $X : \text{Site}$
$f_* : D^b(X, k) \rightarrow D^b(Y, k)$	Derived pushforward along $f : X \rightarrow Y$ where $X, Y : \text{Site}$
$f^* : D^b(Y, k) \rightarrow D^b(X, k)$	Derived pullback along $f : X \rightarrow Y$ where $X, Y : \text{Site}$
$\otimes_k : D^b(X, k) \times D^b(X, k) \rightarrow D^b(X, k)$	Derived tensor product for $X : \text{Site}$
$\mathcal{H}om : D^b(X, k)^{\text{op}} \times D^b(X, k) \rightarrow D^b(X, k)$	Derived sheaf Hom for $X : \text{Site}$
$\mathcal{E}nd : D^b(X, k) \rightarrow D^b(X, k)$	Derived sheaf End for $X : \text{Site}$

where the derived functors are assumed to preserve boundedness.

1.2 Motivations

1.2.1. Monodromy groups are interesting mathematical objects that arise naturally in both geometric and arithmetic contexts. Controlling and understanding monodromy groups has been a central theme in both of these worlds, with significant applications.

1.2.2. On the geometric side, investigation of monodromy goes back to Riemann.

1.2.3 A Question in Complex Geometry. Here is a

1.2.4. On the arithmetic side,

1.2.5 A Question in Number Theory. Here is a

1.3 Structure of the Write-up

1.3.1. In § 2

1.3.2. In § 3

1.3.3. In § 4

1.3.4. In § 5

1.3.5. In § 6

1.4 Acknowledgements

1.4.1. The author thanks his advisor, Prof. Chris Hall, for his guidance and advice.

1.4.2 Declaration of AI Use. The use of generative AI in the preparation of this write-up was limited to the following purposes: polishing wording, phrasing, spelling, grammar, and formatting; offering linguistic advice; answering mathematical questions from the author to help him learn the subject; searching and citing appropriate references; translating reading materials in French into English; retypesetting old references with archaic typesetting for the author’s sanity; providing emotional support for the author; providing fitness advice for the author to stay healthy and energetic.

1.4.3 Typesetting. The typesetting style of this write-up is inspired by [Kat88].

2 Local Systems and Lisse Sheaves

2.0.1 Notation. (Complex Geometric Setting) Define the following collection of notations

$X : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	A smooth projective connected curve over $\mathbb{C}$
$S : \text{Set}^{\text{fin}}$	A finite nonempty subset of closed points of X
$U : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	The open subvariety $X \setminus S$
$j : U \rightarrow X$	The inclusion map
$m : \text{Nat}$	The cardinality of S

3 Perverse Sheaves and the Six Functor Formalism

4 Middle Convolution and Monodromy Groups

5 Mod- ℓ Monodromy of Hyperelliptic Curves

6 Formalization in Lean

References

[Kat88] Nicholas M. Katz. *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, volume 116 of *Annals of Mathematics Studies*. Princeton University Press, 1988.